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Abstract

Geometric algebra (also known as Clifford algebra) is a powerful mathematical tool that offers a natural and direct way to model geometric objects and their transformations. It is gaining growing attention in different research fields as physics, robotics, CAD/CAM and computer graphics. Clifford algebra makes geometric objects (points, lines and planes) into basic elements of computation and defines few universal operators that are applicable to all types of geometric elements. This paper provides an introduction to Clifford algebra elements and operators.

1. Introduction

Many research fields, such as robotics, CAD/CAM, machine vision and computer graphics, rely on geometric models of external world. In these applications geometric objects and their transformations are traditionally formalized using linear algebra [2]. The limitation of the traditional approach is that its primitives are too low-level and all geometric concepts have to be represented by vectors and matrices. This creates a separation between geometric reasoning and matrix-based algorithms, which in turn leads to implementation errors [10]. Geometric algebra (also known as Clifford algebra from its main contributor, W. K. Clifford [1]) is an interesting alternative for its natural and intuitive way to model and manipulate the geometric objects. Compared with traditional linear algebra, geometric algebra introduces primitives (subspaces) of any dimensionality (not only vectors) and defines the operators to make calculations with them. Geometric entities, such as lines, planes and volumes, become basic elements of the algebra and can be manipulated by a rich set of algebraic operators that have a direct geometric

significance [5] - [9]. While linear algebra lacks universal geometric operations, in Clifford algebra the unified treatment of subspaces of any dimensionality makes many operators of the algebra universally applicable to all types of elements. Furthermore, several mathematical theories, such as projective geometry, complex numbers and quaternions, are naturally integrated in geometric algebra providing a unified framework.

This paper gives an introduction to Clifford algebra theory. The rest of the paper is organized as follows. Section 2 introduces geometric algebra subspaces, while in section 3 the Clifford operators are detailed. In section 4 the basic operators of geometric algebra are used to solve simple geometric problems in three-dimensional space, such as reflections and rotations. The models of geometry commonly used in computer science applications, that is the homogeneous model and the conformal model, are described in sections 5 and 6, respectively. Finally section 7 contains conclusions from this work.

2. Clifford algebra elements

This section uses a natural and intuitive way to introduce Clifford algebra basic objects.

2.1. Multiplication of vectors in $\mathbb{R}^2$

Considering an orthonormal basis $\{\mathbf{e}_1, \mathbf{e}_2\}$ of a Euclidean vector space $\mathbb{R}^2$, two generic vectors $\mathbf{a}$ and $\mathbf{b}$ of $\mathbb{R}^2$ can be expressed as linear combinations of the basis elements:

$$\mathbf{a} = a_1\mathbf{e}_1 + a_2\mathbf{e}_2 \quad \mathbf{b} = b_1\mathbf{e}_1 + b_2\mathbf{e}_2$$

Multiplying the two vectors $\mathbf{a}$ and $\mathbf{b}$, it is obtained:

$$\begin{aligned} \mathbf{ab} &= (a_1\mathbf{e}_1 + a_2\mathbf{e}_2)(b_1\mathbf{e}_1 + b_2\mathbf{e}_2) = \\ & a_1b_1\mathbf{e}_1\mathbf{e}_1 + a_1b_2\mathbf{e}_1\mathbf{e}_2 + a_2b_1\mathbf{e}_2\mathbf{e}_1 + a_2b_2\mathbf{e}_2\mathbf{e}_2 \end{aligned} \quad (2.1)$$

By defining the following axioms:

$$\begin{aligned} \mathbf{e}_1\mathbf{e}_1 &= \mathbf{e}_2\mathbf{e}_2 = 1 \\ \mathbf{e}_2\mathbf{e}_1 &= -\mathbf{e}_1\mathbf{e}_2 \end{aligned} \quad (2.2)$$

equation (2.1) can be written as:

$$\mathbf{ab} = (a_1b_1 + a_2b_2) + (a_1b_2 - a_2b_1)\mathbf{e}_1\mathbf{e}_2 \quad (2.3)$$

The product between two vectors results in the sum of a scalar term $(a_1b_1 + a_2b_2)$ and a term $(a_1b_2 - a_2b_1)\mathbf{e}_1\mathbf{e}_2$ composed of a numeric coefficient and $\mathbf{e}_1\mathbf{e}_2$.

How can $\mathbf{e}_1\mathbf{e}_2$ be defined?

Considering two orthogonal vectors ae_1 and be_2 in $\mathbb{R}^2$, their product is abe_1e_2 . The coefficient ab represents the area of the rectangle defined by the two vectors, as depicted in Figure 1. With respect to e_1e_2 , two considerations can be introduced. First, (e_1, e_2) define the plane in which the area lies. Second, the sign inversion in the second axiom of (2.2) can be interpreted by assigning an orientation to the area. Let us explain this concept.

A vector is defined by its magnitude and direction; given a vector a , $-a$ is the vector with the same magnitude, but with opposite direction. Similarly, an orientation can be assigned to an area. $e_2e_1 = -e_1e_2$ is the area element whose orientation is opposite to e_1e_2 , as depicted in Figure 2. Ultimately, e_1e_2 represents the oriented area element in $\mathbb{R}^2$ and it is called a *bivector*. While the vector is a one-dimensional oriented subspace, the bivector is a two-dimensional oriented subspace.

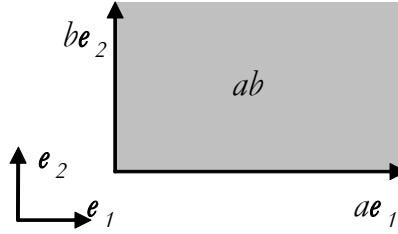


Figure 1. Product of two orthogonal vectors

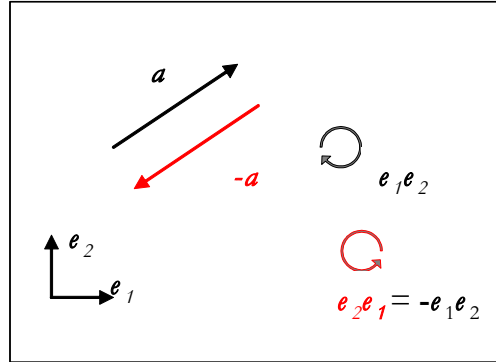


Figure 2. Oriented subspaces

As described in (2.3), the product between two vectors in $\mathbb{R}^2$ results in the sum of a scalar term and a bivector term. If the idea to add heterogeneous elements is accepted, the following expression (obtained by adding a scalar, a vector and a bivector) can be written:

$$a_0 + a_1e_1 + a_2e_2 + a_{12}e_1e_2$$

It represents the most general object that can be obtained combining the basis vectors of $\mathbb{R}^2$ and it is called a *multivector*.

The space containing all the 2D multivectors is called two-dimensional Clifford space (Cl_2) and represents an extension of the Euclidean vector space $\mathbb{R}^2$. The sum and product operations between multivectors are defined in Cl_2 . The algebra based on multivector operations is called Clifford algebra or geometric algebra. The basis elements of Cl_2 are listed in Table 1 along with their grades and multiplicities.

Table 1. Basis elements in Cl_2

Basis element	Grade	Multiplicity
1	0	1
e_1, e_2	1	2
$e_1 e_2$	2	1

The grade of each element indicates the number of basis vectors it contains. The multiplicity of each element represents the number of elements of the same grade contained in the generic multivector. A 2D multivector is a linear combination with real coefficients of the basis elements listed in Table 1.

2.2. Multiplication of vectors in $\mathbb{R}^3$

In section 2.1 the product between two vectors in $\mathbb{R}^2$ has been calculated. In this section the product between two vectors in $\mathbb{R}^3$ is introduced. Considering an orthonormal basis $\{e_1, e_2, e_3\}$ of a Euclidean vector space $\mathbb{R}^3$, two generic vectors $\mathbf{a}$ and $\mathbf{b}$ of $\mathbb{R}^3$ can be expressed as linear combinations of the basis elements:

$$\mathbf{a} = a_1 e_1 + a_2 e_2 + a_3 e_3 \quad \mathbf{b} = b_1 e_1 + b_2 e_2 + b_3 e_3$$

Multiplying the two vectors $\mathbf{a}$ and $\mathbf{b}$, it is obtained:

$$\begin{aligned} \mathbf{ab} &= (a_1 e_1 + a_2 e_2 + a_3 e_3) (b_1 e_1 + b_2 e_2 + b_3 e_3) = \\ &= a_1 b_1 e_1 e_1 + a_1 b_2 e_1 e_2 + a_1 b_3 e_1 e_3 + \\ &+ a_2 b_1 e_2 e_1 + a_2 b_2 e_2 e_2 + a_2 b_3 e_2 e_3 + \\ &+ a_3 b_1 e_3 e_1 + a_3 b_2 e_3 e_2 + a_3 b_3 e_3 e_3 \end{aligned} \quad (2.4)$$

By introducing the following axioms:

$$\begin{aligned} e_1 e_1 &= e_2 e_2 = e_3 e_3 = 1 \\ e_2 e_1 &= -e_1 e_2 \\ e_3 e_1 &= -e_1 e_3 \\ e_3 e_2 &= -e_2 e_3 \end{aligned} \quad (2.5)$$

equation (2.4) can be written as:

$$\begin{aligned} \mathbf{ab} = & (a_1b_1 + a_2b_2 + a_3b_3) + (a_1b_2 - a_2b_1)\mathbf{e}_1\mathbf{e}_2 + \\ & (a_1b_3 - a_3b_1)\mathbf{e}_1\mathbf{e}_3 + (a_2b_3 - a_3b_2)\mathbf{e}_2\mathbf{e}_3 \end{aligned} \quad (2.6)$$

In three dimensions, the product between two vectors also results in the sum of a scalar term and a bivector term. Each of the terms $\mathbf{e}_1\mathbf{e}_2$, $\mathbf{e}_1\mathbf{e}_3$ and $\mathbf{e}_2\mathbf{e}_3$ can be interpreted as the oriented area element which lies on the plane defined by the basis vectors $(\mathbf{e}_1, \mathbf{e}_2)$, $(\mathbf{e}_1, \mathbf{e}_3)$ and $(\mathbf{e}_2, \mathbf{e}_3)$, respectively (see Figure 3). In $\mathbb{R}^3$ another element can be introduced which derives from the product of the three basis vectors: $\mathbf{e}_1\mathbf{e}_2\mathbf{e}_3$.

How can $\mathbf{e}_1\mathbf{e}_2\mathbf{e}_3$ be defined?

$\mathbf{e}_1\mathbf{e}_2\mathbf{e}_3$ represents the oriented volume element in $\mathbb{R}^3$ and it is called a *trivector* (see Figure 3). While the vector is a one-dimensional oriented subspace and the bivector is a two-dimensional oriented subspace, the trivector is a three-dimensional oriented subspace.

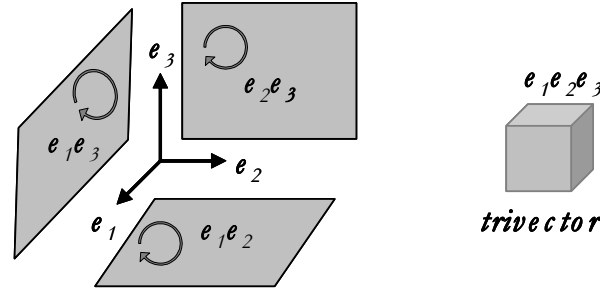


Figure 3. Oriented subspaces in 3D Clifford space

In $\mathbb{R}^3$ the generic multivector can be written as:

$$a_0 + a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + a_3\mathbf{e}_3 + a_{12}\mathbf{e}_1\mathbf{e}_2 + a_{13}\mathbf{e}_1\mathbf{e}_3 + a_{23}\mathbf{e}_2\mathbf{e}_3 + a_{123}\mathbf{e}_1\mathbf{e}_2\mathbf{e}_3$$

The space containing all the 3D multivectors is called three-dimensional Clifford space (Cl_3) and represents an extension of the Euclidean vector space $\mathbb{R}^3$. The basis elements of Cl_3 are listed in Table 2 along with their grades and multiplicities. A 3D multivector is a linear combination with real coefficients of the basis elements listed in Table 2.

Table 2. Basis elements in Cl_3

Basis element	Grade	Multiplicity
1	0	1
$\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$	1	3
$\mathbf{e}_1\mathbf{e}_2, \mathbf{e}_1\mathbf{e}_3, \mathbf{e}_2\mathbf{e}_3$	2	3
$\mathbf{e}_1\mathbf{e}_2\mathbf{e}_3$	3	1

2.3. Subspaces of the n-dimensional Clifford algebra

In this section the concepts intuitively introduced in the sections 2.1 and 2.2 are formalized and generalized in the case of an n-dimensional vector space.

Considering an orthonormal basis $\{e_1, e_2, \dots, e_n\}$ of a Euclidean vector space $\mathbb{R}^n$, the corresponding basis of n-dimensional Clifford space Cl_n consists of all k -dimensional subspaces with $k \leq n$, for all the possible combinations of the n basis vectors, as listed in Table 3.

Table 3. Basis elements in Cl_n

Basis element	Grade	Multiplicity
1	0 (scalar)	$\binom{n}{0}=1$
$e_1, e_2, e_3, \dots, e_n$	1 (vector)	$\binom{n}{1}=n$
$e_1e_2, e_1e_3, \dots, e_1e_n, \dots, e_2e_3, \dots, e_2e_n, \dots, e_{n-1}e_n$	2 (bivector)	$\binom{n}{2} = \frac{n(n-1)}{2}$
$e_1e_2e_3, e_1e_2e_4, \dots, e_1e_2e_n, \dots, e_{n-2}e_{n-1}e_n$	3 (trivector)	$\binom{n}{3} = \frac{n(n-1)(n-2)}{6}$
...	...	...
$e_1e_2e_3 \dots e_n$	n (pseudoscalar)	$\binom{n}{n}=1$

Clifford algebra subspaces are called *blades*: for example, bivectors are blades of *grade 2* (or *2-blades*); trivectors are blades of *grade 3* (or *3-blades*) and so on. According to this formalism, vectors are *1-blades* and scalars are *0-blades*. In Cl_n the total number of these subspaces is 2^n , while the number of k -dimensional subspaces is given by the binomial coefficient

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

It can be observed that in the n-dimensional space the multiplicity of the scalar and the multiplicity of the element of grade n are equal to 1; consequently, the blade of grade n $e_1e_2 \dots e_n$ is called the *pseudoscalar*.

The generic element of the algebra or multivector is a linear combination with real coefficients of the 2^n basis elements listed in Table 3.

3. Clifford algebra operators

This section introduces the basic operators of Clifford algebra: the geometric product, the outer product, the left and right contractions and the unary operators (dual, reverse, conjugate and grade involution).

3.1. The geometric product

Clifford algebra introduces a linear, invertible product called *geometric product* or *Clifford product* that is the fundamental product of the algebra, since other useful products, as the inner product and the outer product, can be derived from it.

Table 4 lists the axioms of the geometric product.

Table 4. Geometric product axioms

$$\begin{array}{ll} e_i e_i = \pm 1 & i = 1, 2, \dots, n \\ e_i e_j = -e_j e_i & i \neq j = 1, 2, \dots, n \\ \lambda e_i = e_i \lambda & i = 1, 2, \dots, n \end{array}$$

The geometric product is an operator for multivectors. Though the rules that are introduced in this section are general, the demonstrations are presented for the two-dimensional case. In the two-dimensional Clifford space Cl_2 , two generic multivectors can be written as:

$$\begin{aligned} \mathbf{a} &= a_0 + a_1 e_1 + a_2 e_2 + a_{12} e_1 e_2 \\ \mathbf{b} &= b_0 + b_1 e_1 + b_2 e_2 + b_{12} e_1 e_2 \end{aligned}$$

The geometric product between two multivectors is performed multiplying each component of an operand with each one of the other operand and simplifying the resulting terms by means of the set of axioms listed in Table 4.

Using the geometric product to multiply $\mathbf{a}$ and $\mathbf{b}$, it is obtained:

$$\begin{aligned} \mathbf{ab} &= (a_0 + a_1 e_1 + a_2 e_2 + a_{12} e_1 e_2)(b_0 + b_1 e_1 + b_2 e_2 + b_{12} e_1 e_2) = \\ &= a_0 b_0 + a_0 b_1 e_1 + a_0 b_2 e_2 + a_0 b_{12} e_1 e_2 + \\ &+ a_1 b_0 e_1 + a_1 b_1 e_1 e_1 + a_1 b_2 e_1 e_2 + a_1 b_{12} e_1 e_1 e_2 + \\ &+ a_2 b_0 e_2 + a_2 b_1 e_2 e_1 + a_2 b_2 e_2 e_2 + a_2 b_{12} e_2 e_1 e_2 + \\ &+ a_{12} b_0 e_1 e_2 + a_{12} b_1 e_1 e_2 e_1 + a_{12} b_2 e_1 e_2 e_2 + a_{12} b_{12} e_1 e_2 e_1 e_2 \end{aligned} \quad (3.1)$$

Studying equation (3.1), it can be observed that the geometric product of two generic multivectors can be expressed as a linear combination of geometric products of basis blades. Therefore, it is sufficient to understand how to calculate geometric products between basis blades.

This section explains the rules to calculate the Clifford algebra products between two arbitrary basis blades. Let A be the index set of basis vectors of the first blade and B the index set of basis vectors of the second blade.

Considering the geometric product between two basis blades, the result contains the basis vectors that are present only in one of the basis blades.

Thus, if an index i is present in both A and B , it is not present in the result, while the result contains the basis vectors whose indices belong to A , but not to B and vice versa.

Returning to equation (3.1), the products between two arbitrary blades can be calculated by a three-step process. First, the scalar values associated to the input blades are multiplied to yield the coefficient of the result. Second, the blade of the result is computed according to the rules described above. Third, the sign of the result depends on the input blades, according to the axioms listed in Table 4. Hence, the sign is the result of three contributions: coefficient signs, algebra signature and blade multiplication. The algebra signature is the pair of integers (p, q) , where p is the number of basis elements which square to $+1$ and q is the number of basis elements which square to -1 .

Finally, equation (3.1) results in

$$\begin{aligned} \mathbf{ab} = & (a_0b_0 + a_1b_1 + a_2b_2 - a_{12}b_{12}) + \\ & (a_0b_1 + a_1b_0 - a_2b_{12} + a_{12}b_2) \mathbf{e}_1 + \\ & (a_0b_2 + a_2b_0 + a_1b_{12} - a_{12}b_1) \mathbf{e}_2 + \\ & (a_0b_{12} + a_{12}b_0 + a_1b_2 - a_2b_1) \mathbf{e}_1\mathbf{e}_2 \end{aligned} \quad (3.2)$$

that is the geometric product of the two multivectors $\mathbf{a}$ and $\mathbf{b}$.

3.2. The other products

The other products of Clifford algebra are the *outer product* ($\wedge$), the *left contraction* ($\lrcorner$) and the *right contraction* ($\llcorner$).

3.2.1. The outer product

The outer product can be interpreted as the opposite of the traditional dot product from linear algebra. While the dot product of two vectors projects a vector onto the other one, the outer product of two vectors extends a vector along the other one. The $\wedge$ (wedge) symbol is used to denote this operator that is also called wedge product. As depicted in Figure 4a, the outer product of two vectors $\mathbf{a}$ and $\mathbf{b}$ results in a two-dimensional subspace or bivector. It has an area equal to the size of the parallelogram spanned by $\mathbf{a}$ and $\mathbf{b}$ and an orientation depicted by the clockwise arc.

The outer product is anticommutative, which means that

$$\mathbf{a} \wedge \mathbf{b} = -\mathbf{b} \wedge \mathbf{a}$$

From this it derives that

$$\mathbf{a} \wedge \mathbf{a} = 0$$

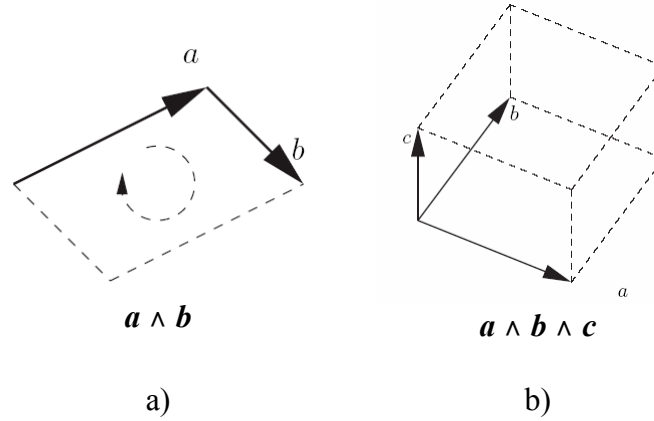


Figure 4. The outer or wedge product

The outer product of two vectors consists in extending a one-dimensional subspace along another to obtain a two-dimensional subspace or bivector. Similarly, if we extend a two-dimensional subspace $a \wedge b$ along a third vector c , we obtain a three-dimensional subspace or trivector, which represents an oriented volume element, as depicted in Figure 4b.

The outer product of two vectors is a blade of grade 2 or a bivector. The outer product of three vectors is a blade of grade 3 or a trivector.

In general, a generic blade of grade k is an outer product of k vectors.

As the geometric product, the outer product is distributive and linear. The outer product of two generic multivectors can be expressed as a linear combination of outer products of basis blades. Therefore, we provide the rules to calculate the outer product between two arbitrary basis blades.

Let A be the index set of basis vectors of the first blade and B the index set of basis vectors of the second blade. The rules to calculate the outer product between the two blades can be expressed as follows: if A and B have a non-empty intersection, the result of the outer product is 0, otherwise the result contains the basis vectors whose indices belong to A , but not to B and vice versa.

3.2.2. The left and right contractions

The left contraction and the right contraction are two of the various versions of the so called *inner product*, which in turn is the generalization of the traditional *dot product* ($\cdot$) defined for vectors in linear algebra. In this paper, we use the left contraction as the specific inner product. In the equations involving only vectors, the left contraction is sometimes written as a dot product.

We use a geometric intuition to explain how the left contraction works. Figure 5 shows how to compute the left contraction between a vector a and a 2-blade B . First, a is orthogonally projected onto B (Figure 5a); then the oriented orthogonal complement with respect to B of the projection is

computed (Figure 5b). Figure 5c shows the orthogonal angle between the projection and the left contraction $\mathbf{a} \rfloor \mathbf{B}$.

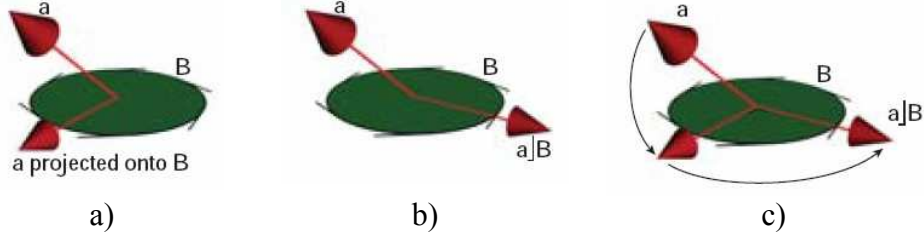


Figure 5. The left contraction [10]

As the geometric product and the outer product, the left and right contractions are distributive and linear. The left (right) contraction of two generic multivectors can be expressed as a linear combination of left (right) contractions of basis blades. Therefore, we provide the rules to calculate the left and right contraction between two generic basis blades.

Let A be the index set of basis vectors of the first blade and B the index set of basis vectors of the second blade.

The result of the left contraction between two basis blades contains the basis vectors whose indices belong to B , but not to A , while the result is 0 if there are elements of A that do not belong to B . Conversely, the result of the right contraction between two basis blades contains the basis vectors whose indices belong to A , but not to B , while the result is 0 if there are elements of B that do not belong to A .

3.2.3. The relation between the algebra products

Let us consider two vectors $\mathbf{a}$ and $\mathbf{b}$ of the two-dimensional Clifford space Cl_2 :

$$\begin{aligned}\mathbf{a} &= a_1\mathbf{e}_1 + a_2\mathbf{e}_2 \\ \mathbf{b} &= b_1\mathbf{e}_1 + b_2\mathbf{e}_2\end{aligned}$$

The geometric product, outer product and inner product of $\mathbf{a}$ and $\mathbf{b}$ can be expressed as:

$$\mathbf{a}\mathbf{b} = (a_1b_1 + a_2b_2) + (a_1b_2 - a_2b_1)\mathbf{e}_1\mathbf{e}_2 \quad (3.3)$$

$$\mathbf{a} \wedge \mathbf{b} = (a_1b_2 - a_2b_1)\mathbf{e}_1\mathbf{e}_2 \quad (3.4)$$

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 \quad (3.5)$$

Equation (3.4) shows that the outer product of two vectors $\mathbf{a}$ and $\mathbf{b}$ is a bivector representing the oriented area of the parallelogram spanned by $\mathbf{a}$ and $\mathbf{b}$ (see Figure 4a).

It is possible to express the outer product and the inner product of two vectors in terms of their geometric product:

$$\mathbf{a} \wedge \mathbf{b} = \frac{1}{2} (\mathbf{ab} - \mathbf{ba}) \quad (3.6)$$

$$\mathbf{a} \cdot \mathbf{b} = \frac{1}{2} (\mathbf{ab} + \mathbf{ba}) \quad (3.7)$$

Equations (3.6) and (3.7) are true only for vectors. From (3.6) and (3.7) the following equation can be derived:

$$\mathbf{ab} = \mathbf{a} \wedge \mathbf{b} + \mathbf{a} \cdot \mathbf{b} \quad (3.8)$$

The geometric product for vectors can be expressed as a combination of the outer and inner products. The outer product is anti-commutative, while the inner product is commutative, which means that:

$$\mathbf{a} \wedge \mathbf{b} = -\mathbf{b} \wedge \mathbf{a} \quad (3.9)$$

$$\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a} \quad (3.10)$$

It is observed that the geometric product is, in general, not commutative ($\mathbf{ab} \neq \mathbf{ba}$), nor is it anti-commutative ($\mathbf{ab} \neq -\mathbf{ba}$). This is a consequence of the fact that the anti-commutative outer product and the commutative dot product are both part of the geometric product.

3.3. The unary operators

3.3.1. Dual

The dual of a generic blade A is defined as:

$$A^* = A \rfloor I_n^{-1} \quad (3.11)$$

where I_n is the blade with respect to which the dualization is performed. I_n is often the pseudoscalar of the space.

The dualization of a blade consists in taking the orthogonal complement of the blade. The dual of A is the part of the space I_n that is not contained in A .

3.3.2. Reverse

The reverse of an arbitrary blade consists in reversing the order of the factors of the blade. For instance, the reverse of a blade $\mathbf{a} \wedge \mathbf{b}$ is

$$(\mathbf{a} \wedge \mathbf{b})^\dagger = \mathbf{b} \wedge \mathbf{a}$$

The reversion only changes the orientation of a blade that is its sign.

In general, the reverse of a blade A_k of grade k is defined as

$$A_k^\dagger = (-1)^{\frac{k(k-1)}{2}} A_k \quad (3.12)$$

3.3.3. Coniugate

The coniugate of a generic blade A is defined as

$$(A_k)_{con} = (-1)^k A_k^\dagger \quad (3.13)$$

3.3.4. Grade involution

The grade involution toggles the orientation of a blade if its grade is odd:

$$\hat{A}_k = (-1)^k A_k \quad (3.14)$$

3.4. Implementation of Clifford blades and operators

How can the basis blades of a Clifford space be represented?

Considering an orthonormal basis $\{e_1, e_2, e_3, \dots, e_n\}$ of a Euclidean vector space R^n , the corresponding n -dimensional Clifford space Cl_n is defined by 2^n basis elements, for all possible combinations of the n basis vectors. To represent them, a n -bit mask can be associated to each basis element, where each bit is associated to a basis vector e_i , $i \in [1, n]$, with e_1 the least significant bit. As example, the blades of Cl_3 and their corresponding bit masks are listed in Table 5.

Table 5. Blade bit masks in Cl_3

Blade	Bit mask		
	e_3	e_2	e_1
1	0	0	0
e_1	0	0	1
e_2	0	1	0
$e_1 e_2$	0	1	1
e_3	1	0	0
$e_1 e_3$	1	0	1
$e_2 e_3$	1	1	0
$e_1 e_2 e_3$	1	1	1

A product operation between two basis blades gives as result another basis blade. According to the rules described in sections 3.1 and 3.2, the following technique can be used to calculate geometric products, outer products and left or right contractions between two basis blades. The bit mask of the result is computed according to the following cases:

- for geometric products, the result bit mask is the XOR of the two input bit masks;
- for outer products, the result bit mask is the XOR of the two input bit masks with the exception that the result will be zeroed if the bitwise AND of the two blades is different from zero.

- for left (right) contractions, the result bit mask is the XOR of the two input bit masks with the exception that the result will be zeroed if the bitwise AND between the first (second) blade and the complemented second (first) blade is different from zero.

4. Geometric transformations

Clifford algebra operators have a very great representational power. As demonstrated in the following sections, the geometric transformations in 3D space, such as reflections and rotations, can be expressed by the algebraic operators introduced in section 3.

4.1. Reflections

Let us suppose to have to reflect a vector $\mathbf{a}$ in a plane with unit-normal $\mathbf{m}$. Let $\mathbf{a}'$ be the reflected vector, as depicted in Figure 6.

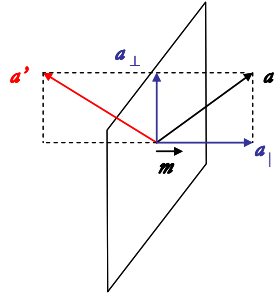


Figure 6. Reflection of a vector $\mathbf{a}$

Pre-multiplying and post-multiplying $\mathbf{a}$ by $\mathbf{m}$ and sign-inverting, it is obtained:

$$-\mathbf{m}\mathbf{a}\mathbf{m}$$

$\mathbf{a}$ can be decomposed in two parts: one part is orthogonal to $\mathbf{m}$ ($\mathbf{a}_\perp$) and the other is collinear with $\mathbf{m}$ ($\mathbf{a}_\parallel$):

$$\mathbf{a} = \mathbf{a}_\perp + \mathbf{a}_\parallel \quad (4.1)$$

Using equation (4.1), the following equation can be written:

$$-\mathbf{m}\mathbf{a}\mathbf{m} = -\mathbf{m}(\mathbf{a}_\perp + \mathbf{a}_\parallel)\mathbf{m} \quad (4.2)$$

and therefore:

$$-\mathbf{m}\mathbf{a}\mathbf{m} = -\mathbf{m}\mathbf{a}_\perp\mathbf{m} - \mathbf{m}\mathbf{a}_\parallel\mathbf{m} \quad (4.3)$$

Since $\mathbf{a}_\perp$ and $\mathbf{m}$ are orthogonal vectors, their product is anti-commutative:

$$\mathbf{a}_\perp \mathbf{m} = - \mathbf{m} \mathbf{a}_\perp \quad (4.4)$$

Conversely, $\mathbf{a}_\parallel$ and $\mathbf{m}$ are collinear vectors; therefore, their product is commutative:

$$\mathbf{a}_\parallel \mathbf{m} = \mathbf{m} \mathbf{a}_\parallel \quad (4.5)$$

Using equations (4.4) and (4.5), equation (4.3) becomes:

$$- \mathbf{m} \mathbf{a} \mathbf{m} = \mathbf{m}^2 \mathbf{a}_\perp - \mathbf{m}^2 \mathbf{a}_\parallel \quad (4.6)$$

Since $\mathbf{m}$ is a unit-vector, $\mathbf{m}^2$ equals 1. Thus equation (4.6) can be rewritten as:

$$- \mathbf{m} \mathbf{a} \mathbf{m} = \mathbf{a}_\perp - \mathbf{a}_\parallel \quad (4.7)$$

$\mathbf{a}_\perp - \mathbf{a}_\parallel$ represents the reflected vector $\mathbf{a}'$. Therefore the reflected vector $\mathbf{a}'$ can be expressed as:

$$\mathbf{a}' = - \mathbf{m} \mathbf{a} \mathbf{m} \quad (4.8)$$

From (4.8) it can be observed that the reflection of a vector $\mathbf{a}$ in a plane with unit-normal $\mathbf{m}$ is obtained by using the geometric product. In fact it is sufficient to pre-multiply and post-multiply $\mathbf{a}$ by $\mathbf{m}$ and sign-invert the result.

4.2. Rotations

Let us suppose to have to rotate a vector $\mathbf{a}$ in the plane generated by the unitary vectors $\mathbf{m}$ and $\mathbf{n}$ (see Figure 7). The rotated vector $\mathbf{a}''$ can be obtained by two reflections: the first one in the plane with unit-normal $\mathbf{m}$ and the second one in the plane with unit-normal $\mathbf{n}$. Recalling equation (4.8), the first reflection gives as result:

$$\mathbf{a}' = - \mathbf{m} \mathbf{a} \mathbf{m}, \quad (4.9)$$

while the second reflection gives as result:

$$\mathbf{a}'' = - \mathbf{n} \mathbf{a}' \mathbf{n} = - \mathbf{n} (- \mathbf{m} \mathbf{a} \mathbf{m}) \mathbf{n} = (\mathbf{n} \mathbf{m}) \mathbf{a} (\mathbf{m} \mathbf{n}) \quad (4.10)$$

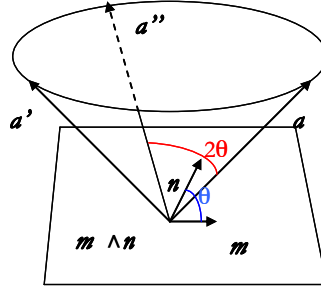


Figure 7. Rotation of a vector a

It can be demonstrated that, if θ is the angle between m and n , the angle between the original vector a and the rotated vector a'' is 2θ . Using the following settings:

$$R = nm \quad \text{and} \quad R^\dagger = mn \quad (4.11)$$

equation (4.10) can be written as:

$$a'' = R a R^\dagger \quad (4.12)$$

where $R^\dagger$ is the *reverse* of R . R is called *rotor*. From equation (4.12) it can be observed that rotations are handled through an interesting “sandwiching product” using geometric products. A rotation of a vector a in the plane defined by two unitary vectors m and n is obtained by sandwiching the vector between the rotor $R = nm$ and its reverse.

Rotors are used to perform rotation of any dimensional subspaces. Thus, equation (4.12), obtained for vectors in 3D space, can be applied to any dimensional subspaces and in any dimensional spaces. Interestingly, equation (4.11) shows that a generic rotor can be derived from the geometric product between two unitary vectors that define both the rotation plane and the rotation angle, where the rotation angle is twice the angle between the two vectors. Since the rotor derives from the geometric product between two vectors, it is a particular multivector, formed by a scalar element and a bivector element.

In three dimensions, a rotor can be expressed as

$$R = (s, b) = q_0 + q_1 e_0 e_1 + q_2 e_0 e_2 + q_3 e_1 e_2.$$

A 3D rotation of a generic vector $v = a_0 e_0 + a_1 e_1 + a_2 e_2$ can be expressed as:

$$v' = R v R^\dagger$$

where v' is the rotated vector and $R^\dagger$ is the reverse of R ,

$$\mathbf{R}^\dagger = (s, -b) = q_0 - q_1 e_0 e_1 - q_2 e_0 e_2 - q_3 e_1 e_2.$$

In three dimensions, rotors are exactly *quaternions*.

5. The homogeneous model

Clifford algebra can be used to implement the most important models of geometry used in computer science applications, such as the homogeneous model and the conformal model. Clifford algebra can represent these standard geometric models increasing their expressive power. In particular, four-dimensional Clifford algebra implements the homogeneous model, while five-dimensional Clifford algebra implements the conformal model, also known as double homogeneous model.

The homogeneous model implemented by 4D Clifford algebra is an extension of the homogeneous coordinates traditionally used to model geometry in three-dimensional space.

The elements of Clifford algebra introduced in the previous sections represent oriented subspaces through the origin. In order to use the elements of the algebra to represent geometric objects, as points, lines and planes, more structured approaches, such as the homogeneous model or the more powerful conformal model, have to be used.

To represent a geometric object, such as a line or a plane, a point on the object is required to define its position in 3D space. For example, a line is described by a direction vector and a position vector, as well as a plane is described by a bivector, which defines its orientation, and a coordinate-vector, which defines its position. To avoid the need for a supporting position, the homogeneous model can be used. The homogeneous model consists in embedding an n -dimensional space in an $n+1$ -dimensional space, adding an extra dimension. Thus, geometric objects from the n -dimensional base space can be represented by pure blades of the $(n+1)$ -dimensional representational space without the need for a supporting position [7].

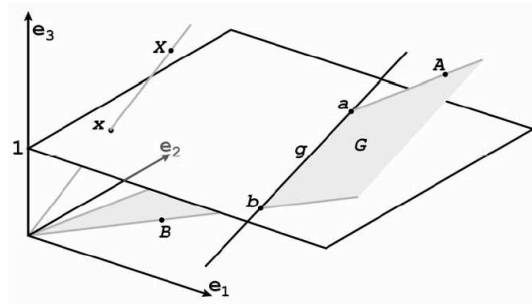


Figure 8. Homogeneous coordinates

By separating the origin of the algebra from the origin of the represented space, the homogeneous model can identify points, lines and planes with pure blades. For example, let us suppose to embed the Euclidean plane (that is the base space) in the three-dimensional homogeneous space (that represents the embedding space), as depicted in Figure 8. The base space, in this case the Euclidean space, is put at a unit distance from the algebraic origin. To represent the line g through points a and b in the Euclidean plane, the direction vector $a-b$ and a supporting coordinate a or b are required. Conversely, in the 3D homogeneous space, it is sufficient to use the bivector $G=a\wedge b$. The bivector represents a plane; the intersection of this plane with the Euclidean plane defines the line g through points a and b (see Figure 8). The bivector $a\wedge b$ defines the line through a and b , while $b\wedge a$ defines the line with opposite direction. Many benefits derive from the use of homogeneous coordinates. If a line has to be rotated, instead of having to rotate two objects (the direction vector and the position vector), it is sufficient to rotate the bivector. Furthermore, the intersection of two lines can be obtained by applying the “meet” operator to two bivectors. As an example, considering two lines in the 3D homogeneous space (the first one through points a and b and the second one through points c and d), the intersection of the two lines can be expressed as the meet operator between bivectors $a\wedge b$ and $c\wedge d$, as depicted in Figure 9.

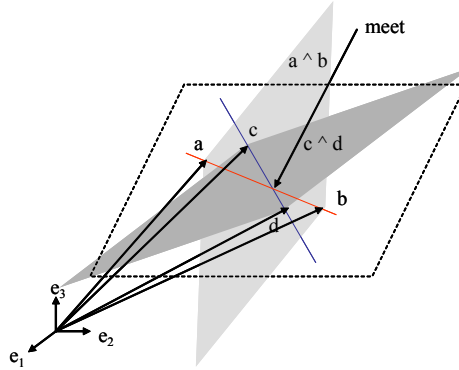


Figure 9. Intersection of two lines

The meet operator between two arbitrary blades A and B is the inner product of the dual of A with B , as described in equation (5.1).

$$A \cap B = A^* \cdot B \quad (5.1)$$

The meet operator gives as result the smallest common subspace of blades A and B . The definition of the dual of an arbitrary multivector A has been given in equation (3.11).

6. The conformal model

Five-dimensional geometric algebra (5D GA), also known as conformal geometric algebra (CGA), implements the conformal model, that is the most powerful and elegant model of 3D Euclidean geometry. The conformal model is also called the double homogeneous model. The homogeneous model consists in embedding an n -dimensional space in an $(n+1)$ -dimensional space, adding one more dimension, which is called the point at origin. The conformal model, on the other hand, consists in embedding an n -dimensional space in an $(n+2)$ -dimensional space, adding two extra dimensions, which are called the point at origin (o) and the point at infinity (∞).

The conformal model is more powerful than the homogeneous model. Compared to the homogeneous model, the conformal model extends the number of geometric objects that can be directly represented by the algebra elements. While homogeneous model can represent only flat geometric objects, the conformal model can represent also round objects, such as circles and spheres, as blades. In the conformal model, all types of geometric objects have a direct association with the blades of the algebra.

Conformal means angle-preserving. In the conformal model, the angle-preserving geometric transformations, such as rotations, reflections, translations, and uniform scaling, can be directly represented by geometric products with five-dimensional rotors.

These are the most basic transformations used in computer graphics, and they would classically be represented with 4×4 matrices, while in the conformal model the natural choice to handle Euclidean transformations is using rotors.

Conformal geometric algebra is a truly coordinate-free model of 3D Euclidean space, in which all geometric operations can be specified without ever referring to an origin.

In the conformal model, geometric objects in the n -dimensional base space can be directly represented through blades of the $(n+2)$ -dimensional embedding space and vice versa blades of the $(n+2)$ -dimensional representational space can be interpreted as geometric objects of the n -dimensional base space. The blade interpretation process in the conformal model is depicted in Figure 10 [10].

First, the blades in the $(n+2)$ -dimensional embedding space are intersected with a hyper-plane. The result of this intersection is a flat in the $(n+1)$ -dimensional space. Then, these flats of the $(n+1)$ -dimensional space are intersected with an n -dimensional paraboloid, the horosphere. The conic resulting from the intersection of the flat with the horosphere is projected onto an n -dimensional hyper-plane orthogonal to the axis of symmetry of the horosphere. These projections represent geometric objects in the n -dimensional space (lines, planes, circles, spheres and so on). In this blade interpretation process two dimensions are lost. So, in the conformal model, to represent an n -dimensional geometry, we have to use an $(n+2)$ -dimensional

geometric algebra. In particular, to model the three-dimensional Euclidean geometry we have to use five-dimensional geometric algebra.

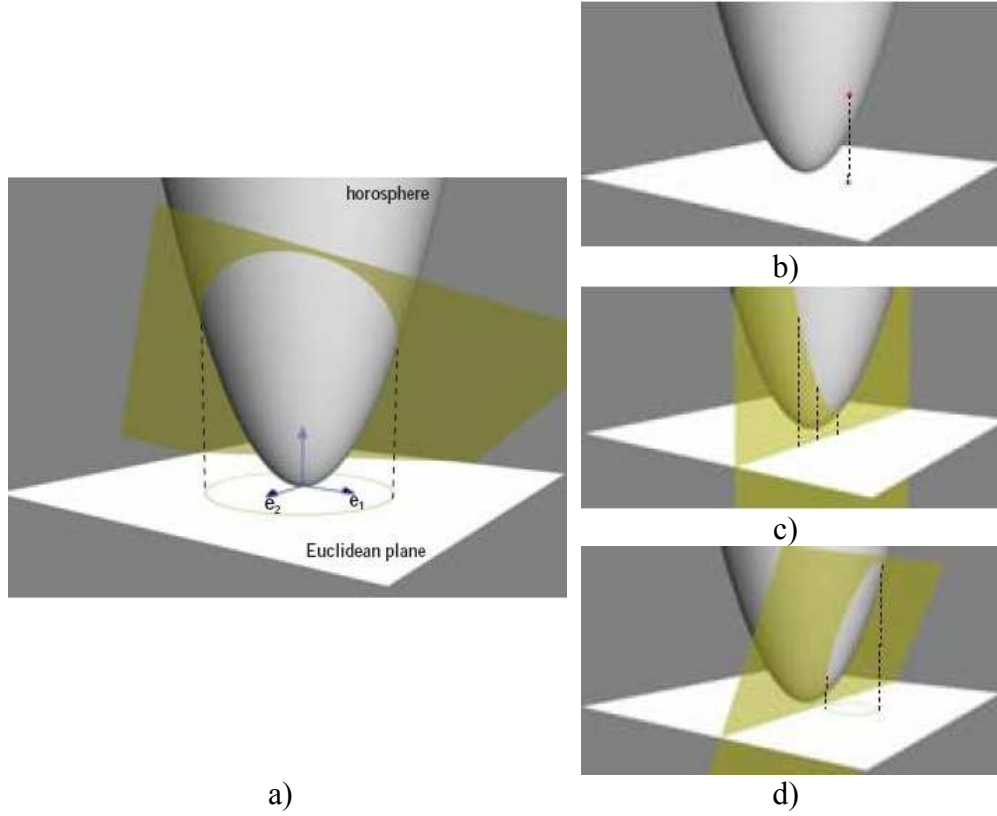


Figure 10. The blade interpretation process in the conformal model [10]

In Figure 10a the blade interpretation process is described in the case of $n = 2$. The figure shows the basis of the conformal space (except the point at origin ϕ). The direction of the point at infinity (∞) is the axis of symmetry of the horosphere. As shown in the figure, the flat in $(n+1)$ -dimensional space is intersected with the horosphere. The resulting conic is projected onto the Euclidean space spanned by the basis vectors e_1 and e_2 . In this case, the result of this projection is a circle.

As depicted in Figure 10b, in the conformal model points are represented by null vectors on the horosphere. Figure 10c shows that lines are represented by planes parallel to the axis of symmetry of the horosphere, while Figure 10d shows that circles are represented by planes which are not parallel to the axis of symmetry of the horosphere.

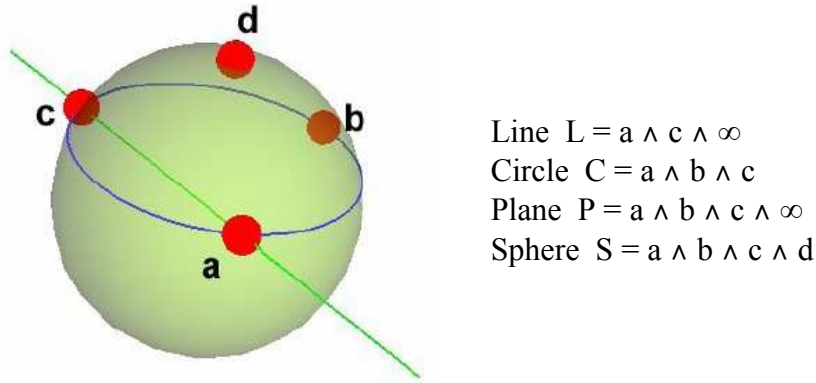


Figure 11. Representation of geometric objects in the conformal model

As shown in Figure 11, a line is created as the outer product of two points on the line and infinity, a circle is the outer product of three points on the circle, a plane is the outer product of three points on the plane and infinity and a sphere is obtained as the outer product of four points on the sphere. Flat subspaces are represented as spheres through infinity.

As shown in Figure 12, the circle deriving from the intersection of two spheres S_1 and S_2 can be obtained as the outer product of the two spheres $S_1 \wedge S_2$.

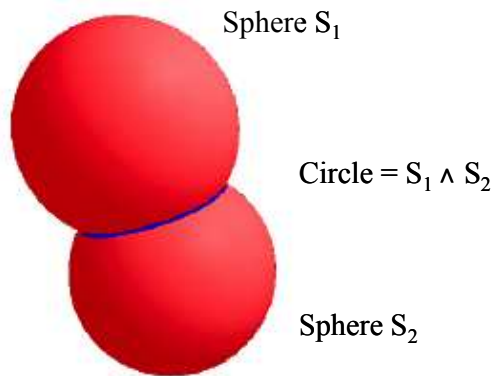


Figure 12. Intersection of two spheres in the conformal model

The conformal geometric algebra is based on the Minkowski metric $R^{n+1,1}$, which is also used in physics for space-time. In particular, the conformal model of the 3D Euclidean space, that is the 5D geometric algebra, is based on the $R^{4,1}$ Minkowski metric.

In general, the notation $R^{p,q}$ means that p is the number of basis vectors which square to $+1$ and q is the number of basis vectors which square to -1 .

For the conformal model of 3-D Euclidean space, we use $R^{4,1}$, to denote that four basis vectors square to 1 and one squares to -1.

Conversely, the notation R^n is used for a space with n positive and no negative dimensions.

The metric of a vector space can be specified by a metric matrix, which is a simple look up table defining the inner product for each combination of basis vectors. In a Euclidean metric, all diagonal metric factors are 1.

For example, the metric matrix for an orthogonal 3D Euclidean basis e_0, e_1, e_2 of R^3 , is the diagonal matrix shown in Table 6.

Table 6. Euclidean metric for 3D space

	e_0	e_1	e_2
e_0	1	0	0
e_1	0	1	0
e_2	0	0	1

According to the Minkowski metric $R^{4,1}$, the metric multiplication table is as shown in Table 7.

Table 7. Minkowsky metric in the conformal model

	e_0	e_1	e_2	e_3	e_4
e_0	1	0	0	0	0
e_1	0	1	0	0	0
e_2	0	0	1	0	0
e_3	0	0	0	1	0
e_4	0	0	0	0	-1

The relation between the basis vectors and the points at origin (o) and at infinity (∞) is as follows:

$$o = \frac{1}{\sqrt{2}} (e_3 + e_4)$$

$$\infty = \frac{1}{\sqrt{2}} (-e_3 + e_4)$$

7. Conclusions

Geometric or Clifford algebra is a new powerful computing paradigm which offers a natural way to model geometric objects independently from their coordinates. Clifford algebra adds to classical linear algebra concepts,

such as scalars and vectors, two-, three-, or higher-dimensional subspaces and defines the operators to perform calculations with them. It is possible, in fact, to add and subtract subspaces of different dimensions, and even multiply them, resulting in powerful expressions that can express any geometric relations and concepts. Geometric objects have a direct representation in geometric algebra and their transformations can be expressed directly by the new powerful operators of the algebra. Clifford algebra is applied to different fields of research as computer graphics, CAD/CAM, robotics, physics and everywhere the description and the manipulation of geometric entities are very important. However, Clifford algebra has been introduced in these fields only in recent years, while the traditional methods have been used for years. Consequently, a lot of work has still to be done to find efficient implementations of Clifford algebra that allows us to benefit from its expressiveness and elegance without sacrificing performance.

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